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Transport

Review on road congestion pricing: a long-term land use effect perspective

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Road congestion pricing (RCP) is an important way of controlling traffic congestion from the perspective of traffic demand management. This paper introduces the concept and four main types of applications of RCP, summarises the development of RCP theory, reveals the first- and second-order effects of RCP on transportation systems and land use, and analyses the long-term impact of RCP on the spatial location structure of land use. Relevant research progress considering land use effects is systematically and comprehensively reviewed from three research perspectives: demand modelling (pre-RCP), monitoring and analysis (post-RCP) and modelling methods (integration models and other models). The results showed that RCP will affect residents' travel behaviour in the short term and will lead to changes in spatial location decisions in the long run, thus creating new travel demands and travel modes, affecting the transportation system and urban spatial patterns again. Implementing cordon charges has an agglomeration effect on the population in city centres. Cordon charges also have a significant negative impact on businesses in the toll area and a dispersive effect on jobs. Combined with appropriate strategies for shared autonomous vehicles, RCP can simultaneously improve transportation efficiency and regional accessibility.

Keywords: land use/policy effect/public policy/review/road congestion pricing/transport management/transportation economy/UN SDG 11: sustainable cities and communities

1. Introduction

Urbanisation is a driving force for the rapid growth of the global economy now and for a long time in the future, but rapid urbanisation has also led to 'urban diseases' such as urban sprawl, high land and house prices, excessive separation of work and home, traffic congestion and environmental degradation in many large cities. The social consequences of traffic congestion are particularly severe, causing substantial economic losses to society, reducing transportation efficiency and accelerating the deterioration of the urban environment. The core of the problem is the imbalance between transportation demand and supply and the spatial and temporal imbalance of traffic flow. Traffic congestion has become a significant obstacle to the sustainable development of urban economies and a bottleneck that determines the success of governments' strategies for better and faster economic development. Solving the problem of traffic congestion in big cities and alleviating the negative impact of traffic congestion on social and economic development are significant issues, related not only to urban economic development but also to people's livelihoods.

In recent decades, road congestion pricing (RCP) has received more and more attention as a direct and economically effective way of dealing with urban traffic congestion. By levying congestion charges, RCP makes travellers aware of the actual social cost of their travel and enables them to use road transport resources more rationally. At present, RCP has been successfully implemented in more than ten countries and cities, including Singapore, London, Oslo and Stockholm (Börjesson and Kristoffersson, 2015; Lehe, 2019; Liu *et al*., 2024).

The implementation of RCP not only affects the short-term behaviour of travellers (e.g. in terms of travel routes, departure times,

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travel mode and travel frequency) and the spatiotemporal distribution of urban road traffic. More importantly, in the long run, RCP will also affect travellers' choice of destination, residence and workplace, and will affect land use elements such as employment and residential distributions, number the of commercial units, business turnovers, land rents and yields in and around the congestion charging area. This change in a city's spatial structure will also lead to changes in the spatial and temporal distribution of residents' travel demand, affecting residents' travel behaviour and road traffic flow, resulting in the old RCP strategy failing to achieve the desired results. Therefore, when formulating a RCP scheme, it is necessary not only to consider its short-term impact on travel behaviours, but also to assess the long-term impacts of RCP on land use in and around the tolled area and to eliminate or mitigate the negative land use effects of RCP, which was the starting point of this study.

The rest of the paper is organised as follows. A brief overview of the concept of RCP and the development of RCP theory is provided in Section 2. The effects of RCP, from the short term to the long term, are analysed in Section 3. The shortcomings of existing studies and an outlook on future research directions are discussed in Section 4, with conclusions presented in Section 5.

2. Overview of RCP

2.1 Concepts and types of RCP

The aim of RCP is to impose tolls on vehicles in specific time periods and on specific road sections, thereby spreading traffic over time and space and reducing congestion during peak hours and on congested road sections. At the same time, RCP induces a shift to high-capacity transportation systems, suppresses the growth of private car traffic, promotes the effective use of small cars, encourages the implementation of multi-person ride-sharing and realises the most effective use of roads to relieve traffic congestion.

RCP schemes that have been applied worldwide can be generally categorised into the following four types (DeCorla-Souza, 2008).

(*a*) Variably priced lanes. This refers to charging different tolls for different roadway lanes, such as express or high-occupancy lanes, one application being the US Interstate I-15 (for high- occupancy lanes).
(*b*) Variable tolls on entire roadways. This strategy not only tolls currently tolled roads and bridges in the road network but also tolls currently free road facilities during peak periods. This is applied in Florida.
(*c*) Cordon charges. Variable or fixed congestion charges are levied on vehicles entering and travelling in predesignated congested areas of the city; applications of this include London and Stockholm.
(*d*) Area-wide charges. In these schemes, the number of miles travelled determines the amount of charge within a defined charging area, and the defined charging area can change dynamically depending on the congestion level. This is applied in Seattle city.

2.2 The evolution of RCP theory

The concept of RCP was first proposed in the 1920s in *The Economics of Welfare* (Pigou, 1920). Pigou pointed out that if a charge is levied on vehicles that choose a road with a shorter distance but a smaller capacity according to the difference between the marginal social cost of the trip and the marginal personal cost of the trip, some vehicles will transfer to the longer distance but larger capacity road, so that the total time of vehicle travel on different roads will be reduced and the welfare of society will be increased – RCP is derived from this idea. The American economist Frank Knight supported Professor Pigou's views (Knight, 1924), laying the fundamental ideological and theoretical foundation for congestion charging from the supply and demand balance perspective and marginal cost pricing theory.

From the 1950s to the 1980s, as the popularity of private cars in many cities led to severe traffic congestion problems, road congestion charging theory also developed. Initially, scholars focused on congestion charging pricing, using supply and demand relationships and marginal costs to study static congestion charging models. For example, Walters (1961) proposed a short-term marginal cost pricing model for the service pricing problem of US highways, studied it quantitatively, and then derived the optimal toll. Scholars then realised that the static RCP model is based on the transportation network under steady-state conditions, while actual travel demand and cost are time-varying. Vickrey (1969) understood traffic congestion as vehicle queuing and assumed that congestion occurs at traffic bottlenecks. Vickrey replaced the cost of queuing time with the charge of RCP as a prerequisite for the equilibrium condition, thus opening up the theory of dynamic tolling. Wie and Tobin (1998) developed two dynamic RCP pricing models based on marginal cost pricing theory and discussed system equilibrium under stable and fluctuating demand.

Beyond the 1980s, RCP policies began to be implemented in various cities worldwide, and RCP theory gradually developed and matured. Although the marginal cost pricing principle is theoretically perfect, it is not ideal in practical applications. This prompted many to study a variety of suboptimal RCPs, which are tolling methods only in some road sections (Liu and McDonald, 1999; Verhoef, 2002). In addition, scholars began to apply bi-level programming models to study RCP policies (Yan and Lam, 1996; Yang and Bell, 1997) or to consider the effects of demand uncertainty.

In recent years, various research methods have been applied to RCP. Cipriani *et al*. (2019) used TransCAD to simulate a traffic model of Rome and obtained results consistent with those from other real-world cases. Geng *et al*. (2023) considered the finite rationality of commuters and used cumulative prospect theory and genetic algorithms to estimate commuters' departure time under uncertain travelling conditions. Various optimisation methods, both gradient-based and gradient-free, have been applied to study

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the optimal RCP policy. Common gradient-based methods include sensitivity analysis (Du and Chen, 2022), stochastic approximation (Gu and Saberi, 2021) and subgradient-based methods (Cao *et al*., 2021). Gradient-free methods such as genetic algorithms (Wei and Sun, 2018) and Bayesian optimisation algorithms (Huo *et al*., 2023) are also frequently used.

As experience of the implementation of RCP policies accumulates, the impact of RCP on urban structure and environmental sustainability has received more and more attention (Feng *et al*., 2023; Li *et al*., 2017; Xi *et al*., 2022) and the concept of integrated planning of land use and transportation has been applied to the study of RCP policy (Zhong and Bushell, 2017a; Zhong *et al*., 2021). Researchers have recently focused on the relationship between RCP and future transportation, particularly the RCP strategy in the context of shared autonomous vehicles (SAVs) (Kaddoura *et al*., 2020; Shafiei *et al*., 2023).

2.3 Effects of RCP

The implementation of RCP policies has several impacts. In cities and countries where RCP has been implemented, the main impacts have been reduced traffic volumes, faster travel speeds, increased public transport use, environmental improvements and increased financial revenues.

Singapore attempted to implement an area licensing scheme as early as 1975. After more than 20 years of operation, the scheme has proven to be an effective means of controlling peak-hour congestion. The number of motor vehicles entering the controlled zone during the morning peak hour has decreased from 74 000 to 41 500 vehicles per day, the average travel speed has increased by 20%, the proportion of people using public transport has increased from 33% to 69% and revenue has also increased (Seik, 2000). Since the introduction of a dynamic electronic road pricing (ERP) system in Singapore in May 1998, traffic volume in the congestion charging zone has decreased by 10–15% and vehicles speeds have increased by almost 30% (Goh, 2002; Kim and Hwang, 2005; Phang and Toh, 1997).

In London, since the introduction of RCP in 2003, the total traffic volume in the congestion area has decreased by 18%, of which the number of cars has decreased by 35% (and subsequent rate adjustments have not changed this trend), the average delay has decreased by 30%, vehicle speed has increased by 17 km/h, the use of public transport has increased by 38% (Prud'Homme and Bocarejo, 2005) and the net benefit of RCP is more than €123 million per year (TfL, 2007).

The Stockholm RCP has reduced traffic through the congested area by 22%, reduced traffic accidents by 5–10% and reduced vehicle emissions in the congested area by 8–14%. However, in the area surrounding the toll area, RCP resulted in a 10% increase in traffic and a maximum 15% increase in travel time. Such negative effects must be thoroughly evaluated and observed over time before researchers reach firm conclusions (Eliasson *et al*., 2009) and policymakers need to define the potential impacts of RCP policies in order to make better plans.

The implementation of RCP policies can significantly impact many levels of travel behaviour, including people's short- and long-term behavioural strategies. In general, the effects of RCP can be categorised into first- and second-order effects, as shown in Table 1.

Table 1 shows that the impacts of RCP implementation are multifaceted, and Figure 1 illustrates the inter-relationship between the long- and short-term effects of RCP. From a short-term perspective, RCP will change residents' travel paths, departure times, travel modes and travel frequency. Studies on the short-term effects after RCP implementation are relatively abundant, and research in this area is more systematic (Abulibdeh, 2022; de Palma and Lindsey, 2011; Guo and Xu, 2016; Huang, 2002; Liu *et al.*, 2014; Naumann *et al*., 2022; Tan *et al.*, 2015; Xiao *et al.*, 2012; Yang *et al*., 2010; Zhang and Yang, 2004; Zheng *et al*., 2023). In the long run, if households or businesses are affected by the RCP policy (i.e. increasing travel or business costs), the locations of residences, businesses and industries, and the location

Table 1. Categorisation of RCP effects (Tillema and van Wee, 2002)

Duration	Impact on residents	Impact on business
First-order effects (changes in the transportation system)		
Short term	Change travel behaviour, travel routes, travel modes, departure/arrival times, ride-sharing, reduce frequency of trips, cancel trips and so on	Companies: Transportation subsidies for employees/customers, adoption of more flexible work schedules Stores: Offer delivery services or online shopping Carriers: Reduce shipment frequency, increase inventory, change modes, change routes, change departure/arrival times
Second-order effects (changes in spatial location of land use)		
Long term	Residents often choose to change spatial location when they are unwilling to accept the losses caused by RCP. Changes in spatial location can be categorised as either a change in the location of the activity or a change in the location of the home	Like residents, firms will choose to relocate if dissatisfied with first-order effects or unwilling to accept losses due to RCP

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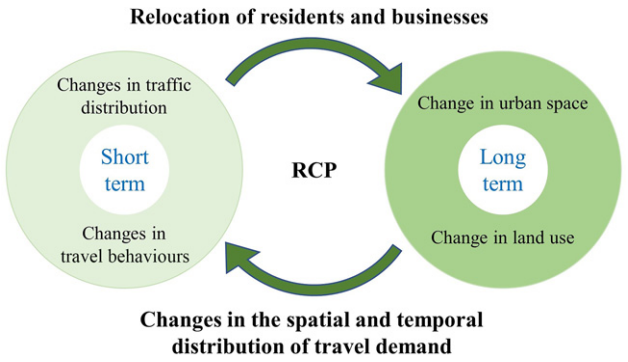

Figure 1. Inter-relationship of long- and short-term effects of RCP

decisions of residents (such as where they live or work) will also be affected. The spatial location decisions of residents and businesses are crucial when studying the effects of RCP because RCP affects the spatial distribution of different land use types in the city, alters travel patterns and mode choices, and ultimately changes traffic flow densities and congestion levels on the transportation network (Zhong *et al*., 2015). Through monitoring and analysis after implementation, it was found that almost 8% of commuters in the City of London changed their destination after the introduction of RCP. For visits to friends and family and shopping and leisure trips, the percentages of trip changes were even higher at 11% and 23%, respectively (TfL, 2007). It is therefore essential to consider the short- and long-term effects of RCP policies and minimise or eliminate negative effects.

At present, scholars have two different views on the possible long-term impacts of RCP on the spatial structure of land use. One point of view is that existing low-priced road transportation facilities make the city present a low-density land use development pattern, and RCP can contribute to a more compact spatial structure of land use in the toll-affected area. The other point of view is that RCP will increase the cost of road use in the toll-affected area and reduce the attractiveness of the toll-affected area, which will have a decentralising effect on the spatial structure of land use in the toll-affected area. Unfortunately, neither of these two conflicting views has enough credible evidence to negate the other. In addition, it is challenging to analyse the impact of RCP on land use since the dominant factors act in opposite directions (Eliasson and Mattsson, 2001).

3. Research on long-term effects of RCP on land use

To avoid the potential long-term negative effects of RCP policies on land use, the development of RCP policies and programmes must be achieved by considering land use and the transportation system together. Existing studies on RCP focus more on short-term effects – that is, the effects on travel distances, departure times, travel modes, travel frequency and other travel behaviours of residents. This type of research assumes that travellers' origins and destinations are fixed. However, this approach ignores the important fact that, in the long run, RCP implementation not only affects the travel behaviour of residents but also affects the accessibility of the congestion pricing area and the cost of travel, which in turn affects the choice of residence and workplace for residents in and around the congestion pricing area, as well as the choice of location for enterprises or businesses, which affects the land use of the area. The spatial separation of different types of land use in cities is the root cause of transportation demand. Therefore, to study the impact of RCP on the transportation system, it is necessary to first study the impact of RCP on land use in areas with different characteristics.

There are few reported studies on the long-term effects of RCP policies and even fewer studies on RCP models and methods considering effects on land use. The existing studies can be grouped into the three research methodologies – demand modelling (pre-RCP), monitoring and analysis (post-RCP) and the modelling method (integration models and other models). The remainder of this section reviews the literature from the perspective of these three approaches.

3.1 Demand modelling

Demand modelling is typically conducted before RCP implementation, using methods such as stated preference surveys or in-depth interviews to examine the likely impacts of RCP implementation. To date, relatively few studies have been conducted in this area and focusing primarily on the impacts of RCP on housing, population, businesses and jobs.

3.1.1 Impact on housing and jobs

Tillema *et al.* (2010a) investigated the impact of RCP on residential (re)location using an intentional choice survey based on the approach shown in Figure 2. They found that 5% of respondents had a high likelihood of relocating their residence if RCP was imposed, compared with 13.5% stating a high likelihood of relocating their job. Further research found that respondents were more willing to pay higher housing costs or accept longer commuting times than they were willing to pay high travel costs. In addition, respondents valued RCP costs more than fuel costs, although the difference between the two was slight (Tillema *et al.*, 2010b).

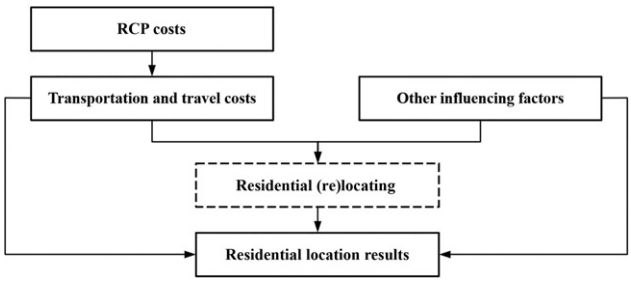

Figure 2. Impact of RCP on location selection of residents

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Arentze and Timmermans (2007) also used an intentional choice experiment and found that about 11.9% of respondents would change their job or home location because of tolls (2.1% likelihood of changing job location and 9.8% likelihood of changing home location). Andani *et al*. (2021) investigated the impact of changes in toll road service levels, socio-economic attributes and latent factors on residential location, route and mode choices of Indonesian residents simultaneously, based on the joint discrete choice model proposed by Bhat and Guo (2007) (Figure 3). Andani *et al*. (2021) found that higher-income individuals were willing to pay more to reside within a 5 km radius of a toll road to gain travel time benefits. On the other hand, those with lower income stated a preference to live farther away, beyond 10 km of a toll road.

3.1.2 Impact on business

Whitehead (2002) used in-depth interviews to examine critical groups (including the UK government, property developers, businesses and academics) and found that the impact of RCP on businesses depended on different causal chains and was, therefore, influenced by different city sizes and levels of regional competitiveness. Through a survey study, Whitehead (2005) also found that firms operating in a marginal revenue state and new firms were most affected by RCP. Saffarzadeh *et al.* (2021) studied the temporary and permanent effects of RCP on customer behaviour in commercial areas of Tehran metropolis, based on the logical approach shown in Figure 4, using an intentional choice survey. They found that customers in three sectors (clothing, electronics and home appliances) were more sensitive to toll changes. In the long run, a 20% increase in toll would lead to the loss of a large number of customers in these three businesses.

Geng *et al*. (2021) conducted in-depth face-to-face interviews in six wholesale markets in Beijing to collect information on the impact of RCP on the relocation strategies of traders under

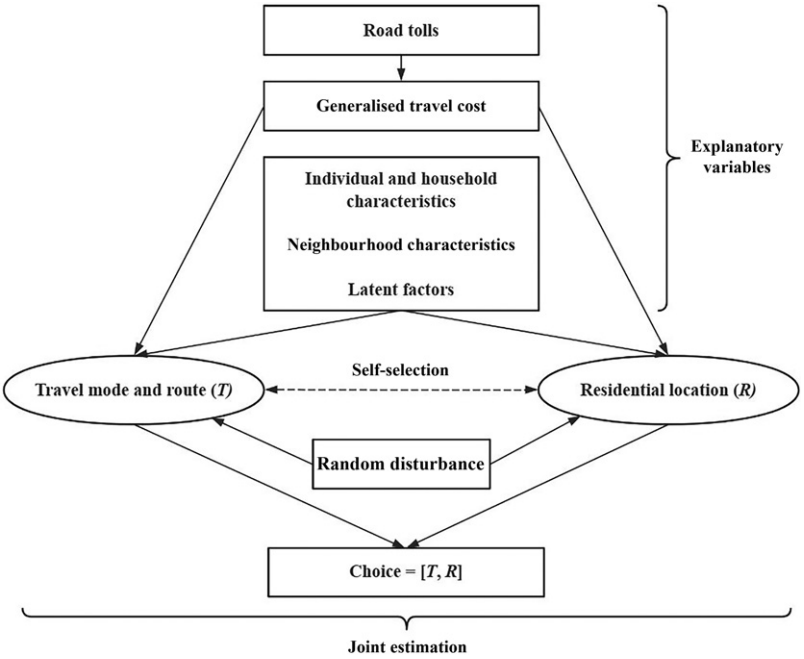

Figure 3. Joint discrete choice model estimation framework

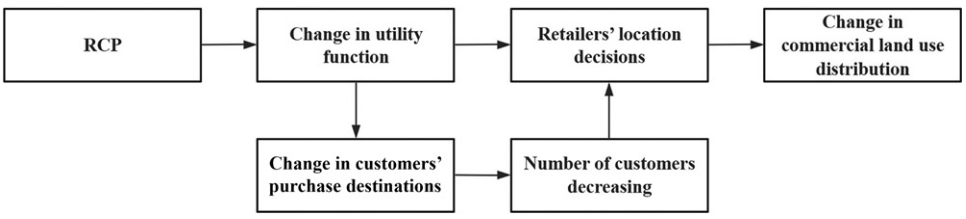

Figure 4. Impact of RCP on commercial land selection

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Beijing's 'non-capital functions relieving strategy'. The study revealed that 35.5% of respondents wanted to leave Beijing for another province, while 39.3% had not decided. In addition, 25.2% of traders expressed a desire to reside in Beijing. Of those, 10% would change jobs, 9.4% would relocate to a different market and 5.9% would change to online stores. Over 60% of respondents wanted the government to designate a new market to preserve the old market's brand and secure sales. Furthermore, over 90% of traders required at least six months to relocate.

3.2 Monitoring and analysis after RCP implementation

Using actual monitoring data to analyse the impacts of RCP implementation has been successfully implemented in Singapore, London, Oslo, Stockholm and other cities (Hess and Börjesson, 2019). Scholars have evaluated the local short-term impacts of RCP implementation using various methods (mostly real-time monitoring, questionnaires, attitudinal surveys etc.), focusing mainly on traffic operating characteristics, individual and household travel behaviour, public transport and the environment (Burt *et al*., 2010; Santos and Fraser, 2006). After introduction of an RCP system, It takes a considerable period of time to fully realise its effects on land use. Empirical studies on the impact of RCP on the type, intensity and employment distribution of different land uses in cities are scarce and observations in this regard are limited, focusing mainly on the impacts on commerce, retail, real estate and employment.

3.2.1 Impact on retail

Quddus *et al*. (2007a) used an econometric model to investigate the impact of RCP on retail sales in central London. A log-linear model with a first-order autoregressive error term was found to model the relationship between sales and congestion charges better than a linear model (Quddus *et al*., 2007b):

1. $$\ln y_t = \alpha + \boldsymbol{\beta} \ln \boldsymbol{X}_t + \boldsymbol{\theta}\, \boldsymbol{D}_t + \varepsilon_t$$

where the error satisfies:

2. $$\varepsilon_t = \rho \varepsilon_{t-1} + \eta_t$$

Here, y_t is sales in period t, $\boldsymbol{X}_t$ is a $k \times 1$ vector of continuous explanatory variables, $\boldsymbol{D}_t$ is an $m \times 1$ vector of dummy explanatory variables, ε is white noise, $\rho\,(-1 < \rho < 1)$ is the autocorrelation coefficient, η_t is an independently and identically distributed error term with zero mean and variance σ^2, α is a constant term to be estimated and $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ are the vectors of parameters to be estimated.

A comparison of six retailers located inside and outside the congestion charge zone revealed that RCP had a significant negative impact on sales inside the congestion charge zone, with one Oxford Street store experiencing an 8% drop in sales. The results of a survey conducted by the Royal Institute of Chartered Surveyors (RICS) confirmed that 90% of retailers inside London's congestion charge zone experienced a drop in sales (RICS, 2005).

The study conducted in London examined the impact of congestion charging on commerce and retail. However, the data and results obtained from post-RCP monitoring and analysis may not be reliable and objective due to external factors such as the global economic crisis and the credit crunch (DeCorla-Souza, 2008).

3.2.2 Impact on real estate

Tang (2021) collected data for the period 2000–2010 on average annual daily traffic flow and housing transactions in the City of London from the Department for Transport and the Land Registry, respectively. Based on hedonic regressions, Tang estimated the impact of traffic on house prices. The formula is:

3. $$Y_{ijt} = \beta^{\mathrm{OLS}} T_{ijt} + X_i'\phi + V_{ji}'\omega + \tau_t + \varepsilon_{ijt}$$

where Y_{ijt} is the logarithm of the price of property i sold in neighbourhood j at time t, T_{ijt} is the logarithm of local traffic conditions as measured by local traffic volume in the vicinity of time t, β^{OLS} is the percentage change in house price associated with a 1% change in local traffic volume, X_i' is a time-invariant property-specific characteristic (e.g. number of bedrooms, garage), V_{ji}' is a time-varying neighbourhood characteristic (e.g. crime or unemployment), τ_t is a time-varying fixed effect and ε_{ijt} denotes both the time-varying and time-invariant error ($\varepsilon_{ijt} = \alpha_i + \theta_{ijt}$).

Calculations showed that, compared with neighbourhoods outside the congestion zone, the implementation of RCP in London resulted in an 8.77% reduction in traffic (1562 fewer vehicles per day). Additionally, house prices in the area increased by approximately 2.84% (£18 555) due to the traffic improvement.

However, Agarwal *et al*. (2015) reached a different conclusion based on implementation of the ERP in Singapore. Their findings, based on real estate transaction data and ERP data from 2007 to 2013, indicate that the ERP rate hike on 1 November 2010 led to a nearly 19% decrease in transaction prices of retail real estate located within the ERP blocked area compared with comparable retail real estate prices outside the ERP blocked area.

3.3 Model method

Studies on the impact of RCP on land use through mathematical models can be divided into integrated model methods and other model methods.

3.3.1 Integrated models

The integrated method refers to applying land use and transportation integration models to analyse the effects of RCP on land use.

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Since the early 1960s, when Wingo (1961) and Alonso (1964) proposed the prototype of integration modelling theory, many integration models have been developed (Abraham and Hunt, 2003; Anas and Arnott, 1994; Putnam, 1983; Echenique *et al*., 1990; Landis, 1994; Mackett, 1983; Martinez, 1992; Salvini and Miller, 2005; Simmonds, 1999; Waddell *et al*., 2003; Wegener, 1982). Since the mid-1950s, some scholars have conducted theoretical research on land models based on location characterisation, including the concentric circle theory and the sector theory proposed by the Chicago School. These models can be classified into the following three types according to the principles on which they are based: (*a*) land use and transportation integration models based on spatial action/gravity models, (*b*) models based on economics and (*c*) models based on microsimulation.

Compared with post-RCP monitoring and analysis, the integrated method has advantages in that it can analyse a more extended period and can exclude the influence of other external factors. In addition, the integrated method does not require actual monitoring data after RCP implementation but only requires the use of traditional land use and traffic survey data to correct the integrated model, based on which the impact of RCP implementation on future land use can be analysed in conjunction with the proposed RCP scenario. Table 2 shows the effects of RCP (borderline tolls and in-area tolls) on land use using the integrated model approach. As shown in the table, most of the integrated model results show that RCP has a clustering effect on the downtown population and a dispersing effect on jobs.

An integrated model can analyse the impact of RCP on land use and other land use or transportation policies on urban land use and transportation. Among them, the Tranus model developed by Delabarra *et al.* (1984), as a representative of regional economic models, has been widely used worldwide. It is theoretically based on random utility theory, bidding rent theory and spatial input–output modelling. Tranus comprises five sub-models – activity location, trip generation, traffic distribution, modal segmentation and traffic allocation. An important feature of the model is the relatively advanced selection and allocation of multi-modal transport paths, which is far superior to the traditional equilibrium allocation of private/public transport.

In recent years, scholars have started to examine the effects of RCP in the future transportation environment (Tscharaktschiew and Evangelinos, 2019; Yang *et al.*, 2019). Zhong *et al*. (2020) identified the combined effects of SAVs and RCP on regional job accessibility through scenario planning, and the results obtained by applying the Tranus model indicated that the introduction of SAVs would promote employment and population development in toll areas and reduce the negative effects of RCP. Liu *et al*. (2024) developed a bi-level planning model with multi-objective functions and multi-decision variables to solve the joint optimal

Table 2. Uses of integrated models to analyse the impact of RCP on land use (cordon charges and area-wide charges)

City and scheme	Reference	Land use and transportation integration model	Main effects of RCP
Broussard: urban zone-wide cordon charges	Lobe and Duchâteau (1998)	Tranus	Change in population within the fee area ranging from −2.0% to +4.0% depending on the fee programme; change in tertiary jobs ranging from −8.0% to +1.0%
London: inner-city cordon charges	CEBR (2001)	LTM/LRM	Commercial land rents and agglomeration increased slightly (0.1–0.5%); effect on residential land use not studied
Oslo: inner-city cordon charges	Jensen (2001)	Tram	Residential areas slightly clustered (0.2%) and jobs dispersed (−0.5%)
Edinburgh: cordon charges (double borderline)	TIE (2004)	Delta/Tram	Population increased in both central city and suburbs. Number of jobs increased in the suburbs. Modest impact on commercial rents and economic growth in the central city
Austin: area-wide charges	Gupta *et al*. (2006)	Dram-Empal/TransCAD	Jobs spread outward from downtown Austin (up to 2.2%)
Washington: inner-city cordon charges	Safirova *et al*. (2006)	Lustre/Start/Relu	Number of jobs in downtown fee area decreased. Workers' wages and welfare gains increased. Magnitude of welfare gains closely related to the redistributive mechanism
Los Angeles: area-wide charges	Anas (2020)	Relu-Tran	RCP in downtown Los Angeles caused some jobs to move outwards. Some consumers from Orange and Riverside counties will move to Los Angeles county or other areas to mitigate the impact of tolls
Jiangyin: inner-city cordon charges	Liu *et al*. (2024)	Tranus	When SAVs and RCP are combined, this can optimise regional accessibility and total flow time by 14.0% and 18.5 %, respectively

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pricing problem of SAVs and RCP considering regional accessibility, using the Tranus model as the underlying model. They found that implementing a joint optimal pricing strategy not only reduces traffic congestion and improves transportation network efficiency, but also optimises a city's spatial layout and promotes compact city development. As a result, regional accessibility increased by 14.0% and total mobility time decreased by 18.5%.

3.3.2 Other models

In addition to using land use and transportation integration models, scholars have developed separate models to specifically analyse the impact of RCP on land use. To study the impact of RCP on commuter and migrant travel flows and the distribution of the unemployed, McArthur *et al.* (2012) considered the commuting or migration decisions of travellers and used microsimulation to study the impact of RCP on a regional labour market. They concluded that while the implementation of RCP increased the social welfare of the region, it also created inter-regional inequalities and caused certain regions to experience a decline in the number of jobs. The likelihood of non-employment inequality was found to increase with an increase in the distance between regions, while shorter distances between regions tended to generate significant migration behaviour. Further research suggests that such inter-regional disparities can be mitigated by increasing the infrastructure level within or between regions.

Anas and Xu (1999) improved the original monocentric city land use model, proposed a generalised equilibrium model and found that RCP causes sprawling cities to become more agglomerated regarding employment distribution and population density. De Lara *et al*. (2013) analysed the long-term effects of RCP on land use using a monocentric city model. After implementing congestion pricing, city and average travel distances decreased by 34% and 15%, respectively, compared with the no-toll scenario. RCP brings the distribution of residences, businesses and industries closer to a city's central business district. Anas and Rhee (2006) analysed the effectiveness of the urban growth boundary (UGB) and RCP in curbing urban sprawl. They found that, in the absence of congestion pricing, an overly stringent UGB causes distortions in urban land use, labour and production markets, resulting in 15% reductions in urban land use, labour and production markets and a 15% revenue loss. Compared with UGB, RCP was found to be more effective.

Simoni *et al*. (2019) developed various RCP strategies for future scenarios. They investigated the impacts of these strategies on network conditions and traveller welfare in Austin, TX, using the agent-based simulation model MATSim. Using agent-based models enables the explicit modelling of various transportation demands and traffic factors when analysing mobility scenarios. Carbone *et al.* (2022) used a sequential experimental game to investigate the responses of merchants and consumers who travel by car to changes in RCP. The results showed that the application of RCP resulted in drive-through shoppers and retail activity leaving the downtown area at the time of the charge, and these adjustments are costly.

4. Conclusions

Relevant research on RCP has been reviewed and summarised. The paper began with a brief overview of RCP, including the concept and the four main types of RCP currently applied. The development of the theory of RCP from its origin to the present day was described. The impacts of RCP policy implementation were analysed, highlighting the changes observed in many cities and their impacts on transportation systems and land use in the short and long term. Last but not least, the focus of this paper was to systematically analyse and summarise the research on the effect of RCP on land use and to sort out the long-term effects from the perspective of three research methods, namely demand modelling (pre-RCP), post-RCP monitoring and analysis and the modelling method used (integrated models and other models).

In terms of the long-term effects of RCP, existing studies show that some residents will change their residential location and job choice after the implementation of RCP, which will have a clustering effect on the population in the city centre. At the same time, RCP will have significant negative impacts on businesses in the toll zone, such as loss of customers and decrease in sales, and will have a dispersing effect on jobs. By optimising the pricing strategies of SAVs and RCP together, significant improvements can be made to both traffic flow time and regional accessibility (Zhong *et al*., 2023).

Analysis of existing studies revealed that the current theoretical foundation is still lacking and the research perspective needs to be adjusted. The land use model is the basis of the travel demand model. If the travel demand model is not integrated with the land use model, the resulting travel demand forecast will be incomplete and unreliable (Anas, 2002; Liu *et al.*, 2010). Therefore, an impact assessment of RCP policy should be integrated into the integrated land use and transportation model framework, and the results obtained will be more convincing. Although applications of the integrated modelling approach to study the land use effects of RCP have begun, studies are far from systematic and in depth. Since the urban spatial structures of different countries and regions have many peculiarities and many of the built environment characteristics of the studied cities have specific differences compared with other cities, the results of existing integrated model analysis often cannot be directly applied and it is necessary to conduct further work for universal applicability.

Existing studies are mainly focused on the short-term impacts after the implementation of RCP – that is, the factors affecting the short-term status of the transportation system, such as residents' travel paths, departure times, travel modes, travel frequency and so on. There are few reported studies on the long-term impacts of RCP on land use and urban spatial structure, such as the location selection of residents (or enterprises). The spatial location decisions of residents and businesses will affect the spatial distribution of different land types in the city, which will change travel patterns and mode choices, ultimately forming a new urban transportation pattern. The long-term impact of RCP on land use is thus an essential element that policymakers cannot ignore. In addition,

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existing studies have only analysed the impact of RCP on the overall land use in the region. These studies could not distinguish the differences in the impacts of RCP on land use in different regions, let alone analyse the reasons for the differences, so the negative impacts of RCP on land use in some fee-sensitive regions were weakened or ignored.

Although some studies have reported negative effects of RCP on land use, such as reduced population density and employment opportunities in the toll zone (Zhong *et al*., 2017; Zhong and Bushell, 2017b), few studies have proposed optimisation methods and measures. More importantly, the impact of RCP on land use is not only related to the RCP scheme (toll zone, target, amount etc.) but is also affected by a series of built environment factors, such as land use characteristics, road network characteristics (layout, carrying capacity etc.) and public transportation networks, so the research conclusions obtained from some cities cannot be directly applied to other cities.

As a future research direction, based on solving the above problems, researchers should also pay attention to more realistic traffic situations; for example, how to construct appropriate integration models to accurately identify the changes brought by RCP policies to the traffic system when multiple motor vehicle control strategies are applied in combination (e.g. limits on vehicle numbers). In addition, given the unique geographic conditions, spatial structure and road infrastructure of different cities or regions, more generalised modelling methods should be developed to explore the influencing factors specific to different regions to better assist decision makers in formulating RCP policies. In addition, with the continuous development of new technologies such as vehicle–road coordination and intelligent connected vehicles, the traditional concept of transportation will be changed, and the way implementing RCP and its significance will need to be re-examined and re-planned.

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